Energy saving target exceeded again in 2020

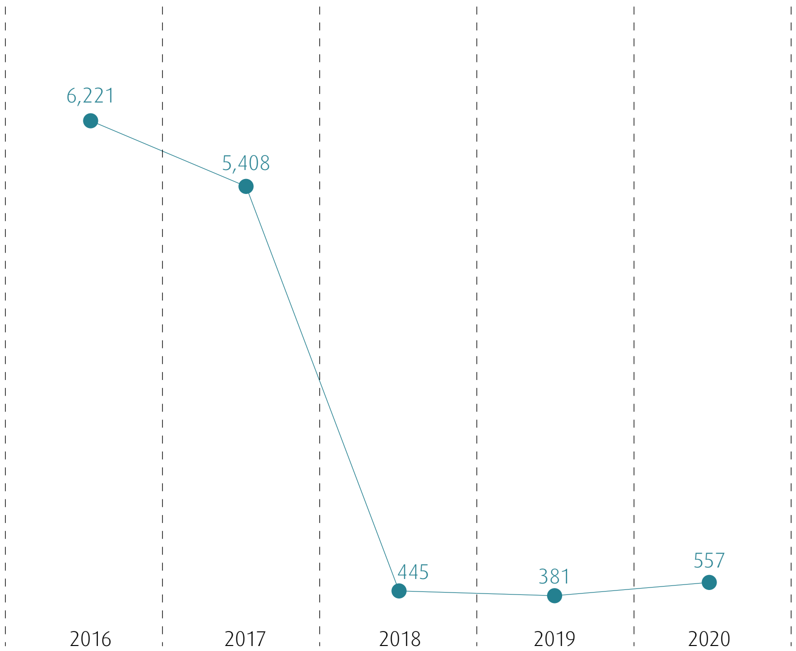
We exceeded our targets once again in 2020 by saving more than 463,600 kWh and thus more than 261 tonnes of CO_2 equivalents. This corresponds to a 4.2 percent reduction. We have set further reduction targets for 2021 and aim to meet these by renewing a further inverter (approx. 200,000 kWh) and deploying energy-efficient filters in our ventilation systems (approx. 60,000 kWh).

Our detailed energy management concept means that we are able to specifically control consumption volumes at our individual sites and plants. We are not simply satisfied with our success to date. Each year, we review our status quo and set new and ambitious energy saving targets. Economic considerations naturally also play a role, as lower consumption volumes enable us to reduce our costs.

As our research and production sites are located in Germany and the UK, our targets mainly refer to these locations. In Asia and the USA, we mainly lease office space and, as lessors, have hardly any scope to introduce energy saving measures.

Wherever possible, AIXTRON will continue to initiate and implement projects to further reduce its energy consumption in future as well. The results confirm the approach we have taken, as is very clearly apparent from the table and chart below.

Total CO_2 equivalent emissions (in t) in Germany



Development in CO_2e emissions at the Herzogenrath site. The 46 percent increase in the 2020 year under report was due a significant rise in sulfur hexafluoride (SF6). To validate a new test, we had to perform several tests in 2020 and also simulated larger-scale leakages. We simulate a gas leakage within our equipment and test how efficiently the ventilation evacuates this gas and how much of this gas escapes into the room and/or users' breathing zones. For this, we require SF6 as this gas does not occur in the natural atmosphere. This is the only way to exclude the possibility of inaccurate measurements. Larger volumes of SF6 were required than originally planned. In this respect, the volume stated for the 2020 year under report represents an exception. Regrettably, no alternatives to SF6 are currently available.

Table with key figures for energy use (Herzogenrath site) Scope 1 + 2

	2017	CO_2e in tonnes	2018	CO_2e in tonnes	2019	CO_2e in tonnes	2020	CO_2e in tonnes
Electricity (kWh)	9,595,253	5,056.7	10,033,811	0	10,658,090	0	10,444,472	0
Natural gas (kWh)	568,181	125	685,610	151	608,462	122.88	858,594	173.44
District heating (kWh)	1,821,360	218.6	2,406,370	289	1,957,130	234.86	1,400,260	268.23
Sulphur hexafluoride (SF6) in kg	---*)	---*)	---*)	---*)	0.772	17.6	4.8	109.4
Nitrogen (N_2) in tonnes	2,050	0	2,290	0	2,612	0	2,616	0
Argon (tonnes)	375	0	23	0	24	0	11	0
Hydrogen (H_2) in m³	10	0	11.3	0	13	0	15.6	0
Water total (in m³)	18,961	7.7	11,833	4.8	13,288	5.4	14,799	6.0
Waste water at cooling towers (in m³)	2,656	0	1,035	0	1,112	0	1,058	0
Total (kWh)/CO_2e in tonnes	**11,984,794**	**5,408**	**13,125,791**	**445**	**13,223,682**	**381**	**12,703,281**	**557**

Consumption of energy and other significant resources at AIXTRON in Herzogenrath. *)
Of perfluorinated and polyfluorinated chemicals, sulfur hexafluoride (SF6) was included in the calculation for the first time in 2019

Key figures for energy use (UK, Asia, USA) Scope 2

2018

	UK	USA	China	Japan	South Korea	Taiwan	∑
Electricity (kWh)	887,727	42,414	26,937	50,942	70,267	73,800	1,152,087
CO_2e in tonnes	0	22.4	14	27	37	39	139.4

2019

	UK	USA	China	Japan	South Korea	Taiwan	∑
Electricity (kWh)	856,253	41,413	27,201	42,160	173,783	75,472	1,216,282
CO_2e in tonnes	0	0	16.94	22.02	93.37	39.41	171.74

2020

	UK	USA	China	Japan	South Korea	Taiwan	∑
Electricity (kWh)	972,946	25,607	22,883	38,531	102,497	65,590	1,228,054
CO_2e in tonnes	0	0	14.33	19.21	55.30	41.07	129.91

Energy consumption at the AIXTRON Group (excluding Germany). The significant reduction in energy consumption in the USA in 2018 was due to the sale of the ALD business and a relocation to new premises. The reduction in CO_2 emissions in the UK and the USA was achieved by converting to green electricity. The substantial increase in South Korea in 2019 was due to the extension of the site by our subsidiary APEVA Korea in 2019.

Promoting biodiversity

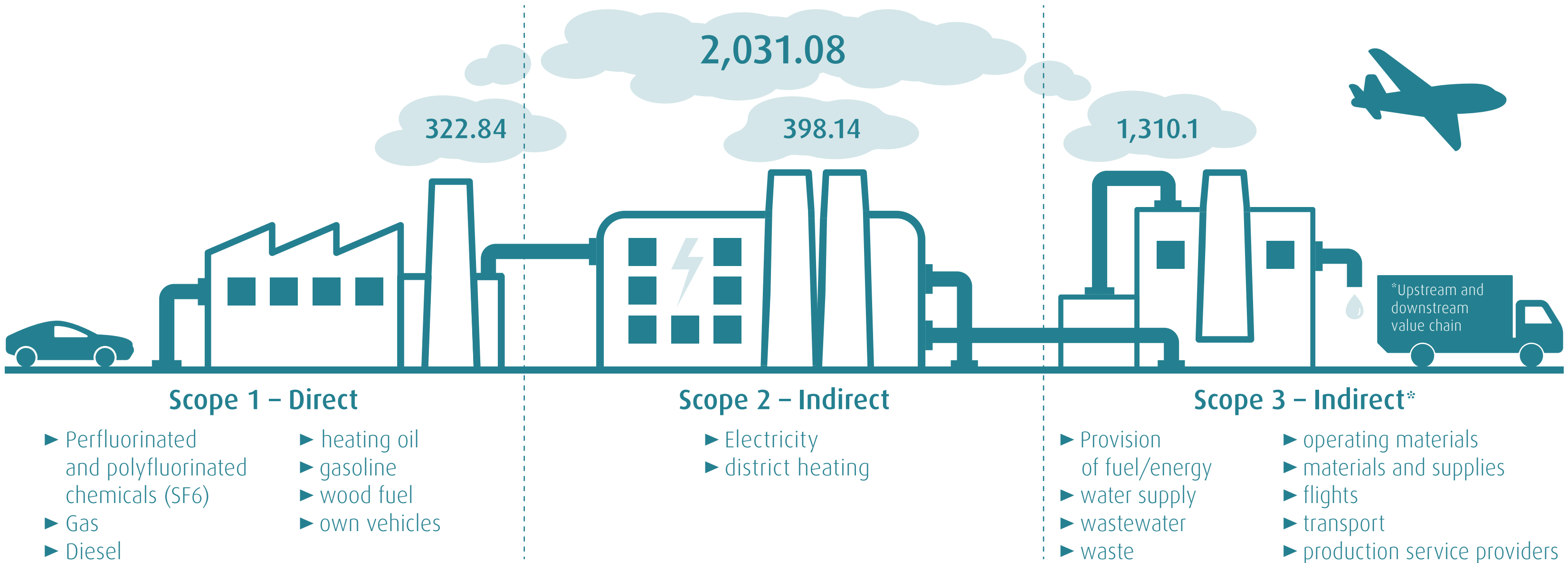
We have converted an area of around 12,500 m² at our company premises into bee and insect pastures. In 2020, we added a “Benjes” hedge and sowed further seed mixtures, including Hönningenbienen seeds. This way, we are supporting the accumulation of wild bees in accordance with the recommendations of the Nature and Biodiversity Conservation Union (NABU).

Building a “Benjes” hedge involves planting several wooden stakes in two rows in the ground and then stacking deadwood, brushwood, and other garden waste to form a hedge. Insects, amphibians, reptiles, and spiders benefit from the deadwood, as do birds, bats, hedgehogs, dormice, and other animals. As a general rule, the thicker the deadwood is, the better suited it is as a habitat and the more nutrition can be found. And deadwood is particularly rich in species if it receives sunshine or is positioned upright. In the summer, there is an endless bustle of insects, such as blue carpenter bees, and lizards and other thermophile species can also be seen.

Climate balance sheet successfully consolidated

Consistent with our aim of continually reducing our CO_2 emissions, since 2018 we have also recorded the key figures for our international subsidiaries around the world. In 2019, we presented a comprehensive balance sheet for the first time for AIXTRON SE and our subsidiary APEVA SE. This balance sheet initially recorded Scope 1 and 2 emissions. With the aim of compiling a complete climate balance sheet including all scope emissions relevant to the GHG Protocol, in 2020 began recording Scope 3 emissions in the company's upstream and downstream business activities as well. These figures will gradually be included in our future reporting and will play a major role in the development of our sustainability targets for the future.

At our customers, our products and equipment help to reduce the volume of resources used compared with older technologies. To account for this contribution to efficiency enhancement in any assessment of our climate balance sheet, we have decided to present developments in relation to our revenues. ■



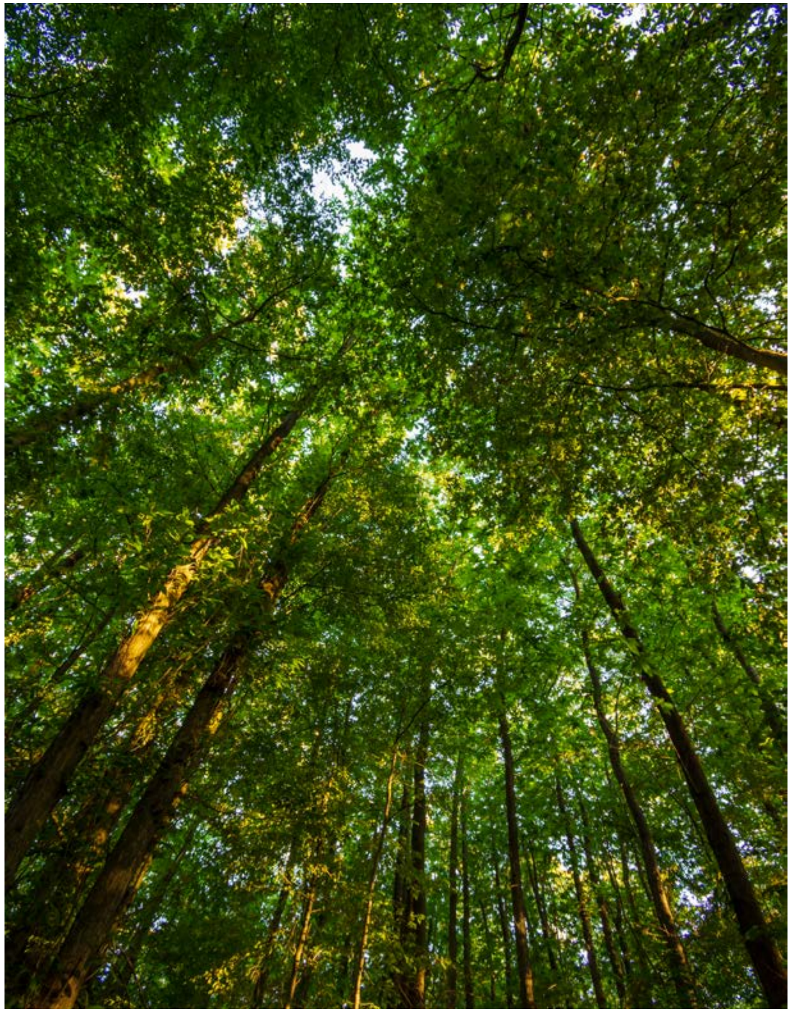
Overview of CO emissions [in metric tons of CO2 equivalents].

CO_2e emissions in absolute figures and in relation to revenues

In total, we emitted 2,031.08 tonnes of CO_2 equivalents in 2020 (2019: 4,215.37 tonnes), which corresponds to a reduction of almost 52 percent. Emissions comprise Scope 1 emissions of 322.84 tonnes (2019: 170.8 tonnes), Scope 2 emissions of 398.14 tonnes (2019: 406.6 tonnes), and Scope 3 emissions of 1,310.1 tonnes (2019: 3,637.97 tonnes) (cf. chart on page 49).

In relation to revenues at AIXTRON SE, emissions fell to 7.54 tonnes per EUR 1 million of revenues (2019: 16.24 tonnes/EUR 1 million).

This reduction is due to the measures taken by the government and resultant restrictions in connection with the coronavirus pandemic, which were reflected in substantially reduced travel activities by our employees. This factor offset the slight increase in Scope 1 emissions.



Offsetting projects in Peru and Uganda

Since 2019, we have offset the unavoidable CO_2 emissions resulting from our business activities, which have been recognized to date in our climate balance sheet, by supporting two certified climate protection projects.

By supporting these two projects, we offset exactly that amount of CO_2 equivalents which corresponds to those emissions from our business activities that we cannot avoid. In selecting the projects, we made sure that the two projects meet the highest certification standards and thus satisfy strict, internationally recognized norms.

Dr. Felix Grawert · Executive Board member

We selected the projects due to their positive impact on the environment, climate, and local populations. The emissions saved are regularly reviewed and confirmed by independent experts.

In the interests of sustainable development, we decided in 2020 already to extend the period for which we are supporting the two projects by a further three years to 2023.

We support certified climate project projects

Protecting rainforests

The sustainable forestry project in the "Madre de Dios" region of Peru works with local residents to implement measures and initiatives to use the Amazon forest on a sustainable basis and to access alternative sources of income for the local population.

Energy-efficient and healthier stoves

In Uganda, we are supporting the distribution of energy-efficient stoves in private households. These enhanced stoves help families to save up

to 50 percent of the fuel they use and thus to reduce the concentration of harmful substances in the air in their kitchens and living rooms. Further details about the projects we support can be found on our website.